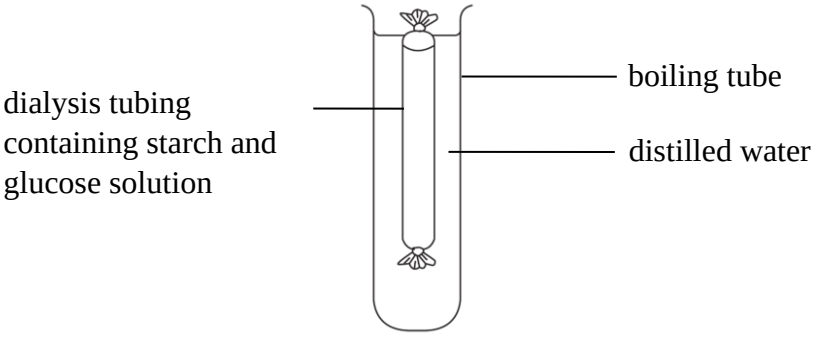
Short questions

|!|BCESB0040000001|!| ★

A model cell is set up as shown below and left for one hour.



Describe the movement of molecules during the hour by

(a) diffusion. (4 marks)

(b) osmosis. (2 marks)

##

(a) The dialysis tubing is differentially permeable. (1)
Glucose molecules are small enough to pass through the pores of the tubing. (1)
They move out of the tubing by diffusion down the concentration gradient. (1)
Starch molecules are too large and cannot pass through the tubing. (1)

(b) The water potential inside the tubing is lower than the distilled water outside. (1)
There is a net movement of water molecules from the distilled water into the tubing by osmosis. (1)

(6 marks)

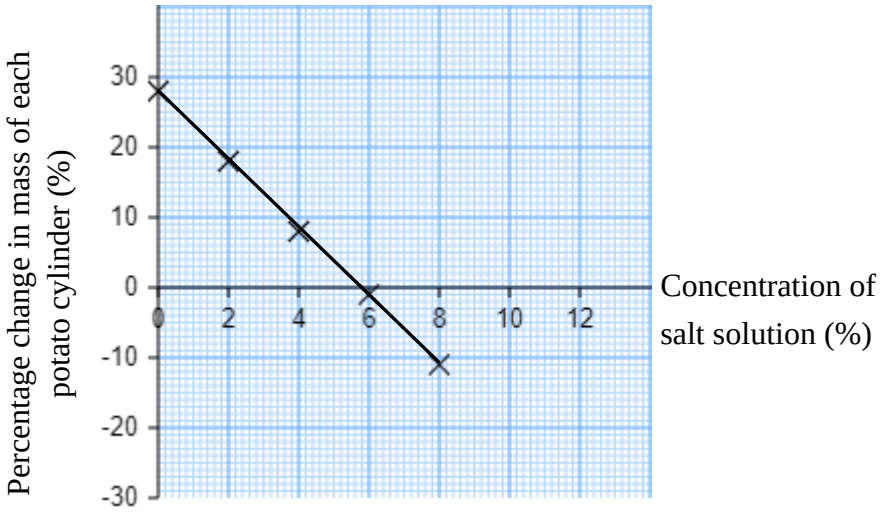
##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)	Applying concepts	2

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|!|BCESA0040000002|!| ★★ **SI**

A student carried out an investigation to determine the water potential of potato tissue. She cut 5 cylinders from a potato. She measured the mass of each cylinder and then placed each one in salt solution (0%, 2%, 4%, 6% and 8%). After 1 hour, she measured the mass of each cylinder again and calculated the percentage change in mass. The results were plotted in the graph below.



(a) Give a title to this graph. (1 mark)

_____

_____

(b) Why did the student cut out the cylinders from the same potato? (1 mark)

_____

_____

(c) At which salt concentration did the mass of the potato cylinder remain unchanged? (1 mark)

_____

(d) Based on the graph, predict the salt concentration that would result in a 30% decrease in mass. (1 mark)

_____

##

(a) The percentage change in mass of potato cylinders in different concentrations of salt solution (or other reasonable answer) (1)

(b) To ensure that the water potential of the cylinders are roughly the same. / The water potential of different potatoes may differ greatly. (1)

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(c) 5.8% (±0.2%) (1)

(d) 12% (1)

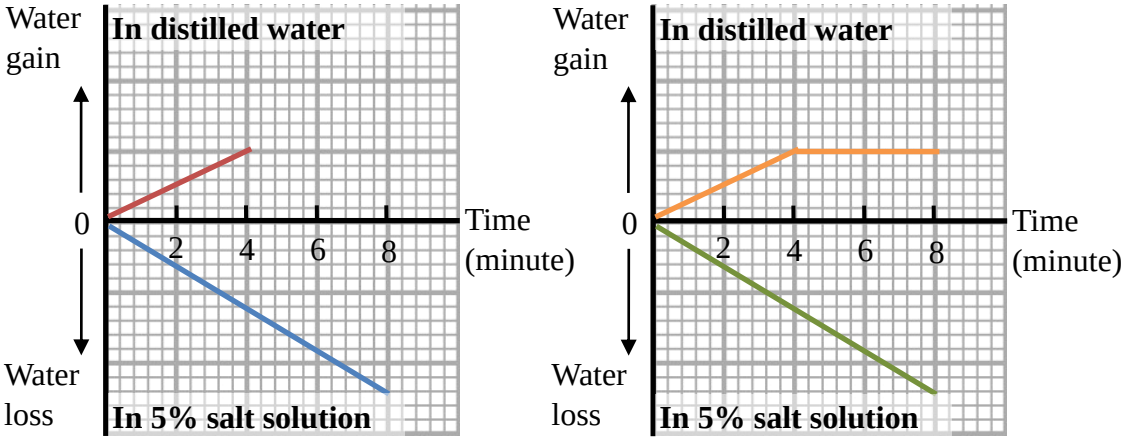
(4 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Practical skills	1
(c)	Analysing data	1
(d)	Analysing data	1

|!|BCESA0040000003|!| ★★

Red blood cells and plant cells were placed in distilled water or 5% salt solution respectively. Their gain or loss of water are shown in the graphs below.



(a) Account for the difference between the changes in the two types of cells in distilled water. (4 marks)

(b) Describe the physical appearances of the two cell types after 8 minutes in 5% salt solution. (2 marks)

##

(a) The water potential of distilled water is higher than that of the cytoplasm of red blood cells. The red blood cells gain water by osmosis (1)
and eventually they burst. (1)
The water potential of distilled water is higher than that of the cell sap of plant cells. The plant cells gain water by osmosis. (1)
The cell wall is rigid to resist the further entry of water after 4 minutes. (1)

(b) The red blood cells shrink and become wrinkled. (1)
The plant cells become plasmolysed. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	2
(b)	Applying concepts	2

|!|BCESB0040000004|!| ★

Different substances can pass through the cell membrane by various methods; diffusion and active transport are two examples.

(a) State ***two*** differences between diffusion and active transport. (2 marks)

(b) Vitamin A and sodium in ingested food can be absorbed through the intestinal walls. Vitamin A is fat-soluble and sodium ion is water-soluble. Explain how the different routes taken by these substances through the cell membrane are determined by the structure of the cell membrane. (4 marks)

##

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(a) Any ***two set*** of the following: (2)

Diffusion	***Active transport***
• does not involve carrier proteins	• involves carrier proteins
• does not require energy	• requires energy
• molecules diffuse down the concentration gradient	• molecules can be moved down or against the concentration gradient

(b) For vitamin A:
It is fat-soluble, so it can dissolve in the phospholipid bilayer of the cell membrane (1)
and pass directly through the phospholipid bilayer by diffusion. (1)
For sodium ion:
Since it is water-soluble, it is repelled by the phospholipid bilayer and can only cross the cell membrane by passing through protein channels (1)
by active transport which requires energy. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	4

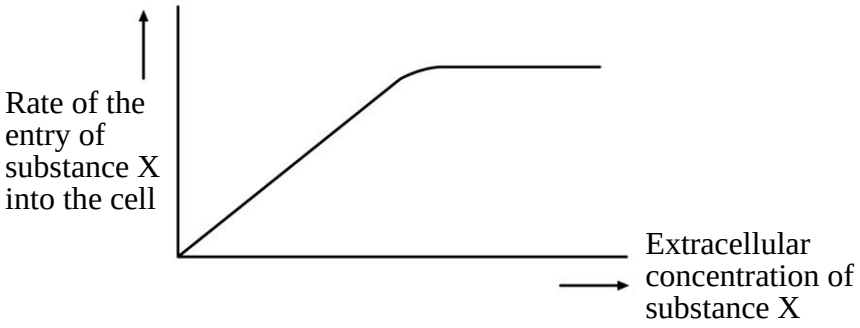
|!|BCESB0040000005|!| ★

Some substances directly diffuse through the phospholipid bilayer of the cell membrane. Some substances diffuse through channel proteins or carrier proteins.

(a) How do you distinguish diffusion through protein channels and active transport? (1 mark)

_____

(b) The following graph shows the relationship between the rate of entry of substance X into a cell and its extracellular concentration.



(i) Based on the graph, deduce whether or not the diffusion of substance X across the cell membrane requires protein channels (2 marks)

(ii) Suggest what substance X may be. (1 mark)

##

(a) Diffusion through protein channels does not consume energy from the cell. Active transport requires energy to change the shape of carrier proteins. (1)

(b) (i) The curve becomes flattened after a certain concentration of substance X is reached (1)
because the channel proteins (or carrier proteins) are saturated with substance X. (1)

(ii) ion / other polar molecule (e.g. amino acid and glucose) (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)(i)	Analysing data	1
	Applying concepts	1
(b)(ii)	Applying concepts	1

|!|BCESC0040000006|!|

Some cells use phagocytosis to take in large particles.

(a) Describe how a cell takes in a large particle by phagocytosis. (3 marks)

(b) Give an example of cell or organism which can carry out phagocytosis. (1 mark)

(c) State the significance of phagocytosis to the cell or organism you answered in (b). (1 mark)

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##

(a) The cell membrane invaginates to form a pit / extends to form pseudopodia (1)
to enclose the particle in a vacuole. (1)
The vacuole detaches from the cell membrane and enters the cytoplasm. (1)

(b) phagocyte / *amoeba* / other reasonable answer (1)

(c) Phagocytes engulf the invading pathogens by phagocytosis to protect the body against pathogens. / *Amoeba* uses phagocytosis for feeding. (1)

(5 marks)

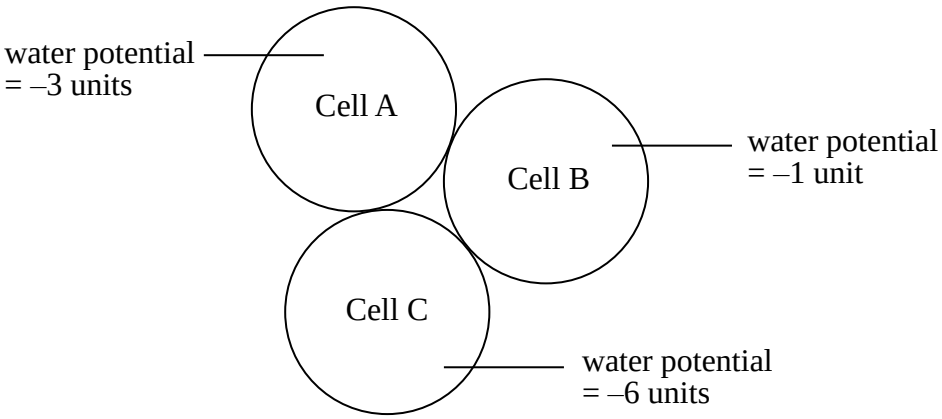
##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESB0040000007|!| ★

(a) What is osmosis? (2 marks)

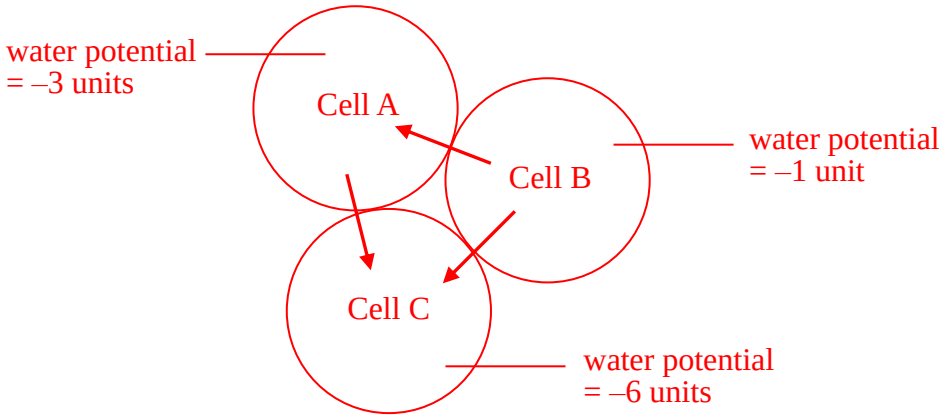
(b) The diagram below shows three cells (A, B and C) which are in contact with each other. The water potential of the three cells is different. Draw arrows in the diagram to show the direction of the net movement of water molecules among these cells. (3 marks)



##

(a) Osmosis is the net movement of water molecules from a region of higher water potential to a region of lower water potential (1)
across a differentially permeable membrane. (1)

(b) 1 mark for each arrow: (3)



(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	3

|!|BCESC0040000008|!|

For each of the transport mechanism listed in column 1, select from column 2 one phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1		***Column 2***
simple diffusion	(a) _____	A. transports water across cell membrane
osmosis	(b) _____	B. transports glucose across cell membrane
active transport	(c) _____	C. transports amino acids across cell wall
		D. transports steroid hormones across cell membrane

##

(a) D (1)
(b) A (1)
(c) B (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

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|!|BCESC0040000009|!|

Fill in the blanks to complete the following paragraph. (6 marks)

Water is very important to life. It enters and leaves living cells by (a)_____. Plant cells have a cell wall made of cellulose. The wall is porous and (b)_____ permeable. In contrast, the cell membrane is mainly made of phospholipids and proteins and is (c)_____ permeable.

Water can move through the cell membrane. The tendency of water molecules to move from one place to another is described as the (d)_____. When a plant cell is immersed in a (e)_____ solution, there is a net intake of water molecules into the cell. As a result, the cytoplasm enlarges until it presses against the cell wall. The cell is described as (f)_____.

##

(a) osmosis (1)
(b) freely / fully (1)
(c) differentially (1)
(d) water potential (1)
(e) hypotonic (1)
(f) turgid (1)

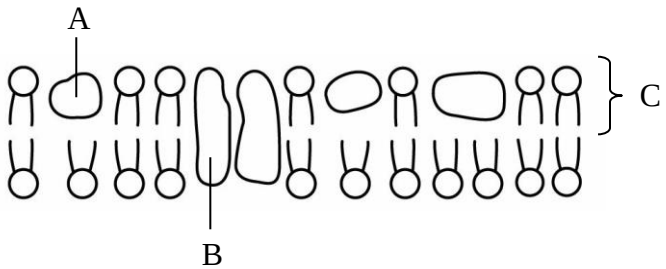
(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

|!|BCESC0040000010|!|

The diagram below shows the fluid mosaic model of the cell membrane.



(a) Suggest a function of structure A. (1 mark)

(b) Name structures B and C. (2 marks)

(c) Explain how the arrangement of structure B in the cell membrane is related to its function. (2 marks)

(d) In the diagram above, draw the path by which an oxygen molecule passes through the cell membrane. (1 mark)

##

(a) enzyme / receptor protein / recognition marker of cells / other reasonable answer (1)

(b) B: channel protein (1)
C: phospholipid (1)

(c) The channel protein spans completely through the cell membrane (1)
and forms a pore through which particular molecules can pass. (1)

(d) path showing that the oxygen molecule diffuses directly through the phospholipid bilayer (1)

(6 marks)

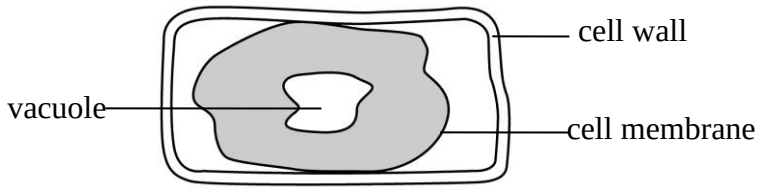
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Analysing data	1
	Applying concepts	1
(d)	Applying concepts	1

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|!|BCESC0040000011|!|

The figure below shows a cell immersed in sucrose solution.



(a) Is the cell shown an animal cell or a plant cell? Explain your answer. (2 marks)

(b) Name the phenomenon shown in the figure above. (1 mark)

(c) Explain the above phenomenon. (3 marks)

##

(a) A plant cell (1)
because it has a cell wall which is absent in animal cells (1)

(b) Plasmolysis (1)

(c) When the plant cell is placed in a hypertonic solution / When the plant cell is placed in a solution with a lower water potential than the cytoplasm of the cell, (1)
there is a net movement of water molecules out of the cell by osmosis. (1)
The cytoplasm shrinks and the cell membrane detaches from the cell wall. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	1
(c)	Applying concepts	3

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